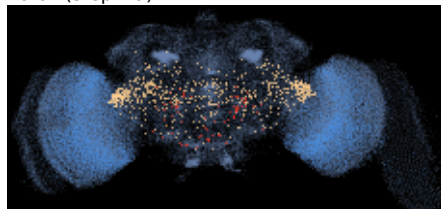


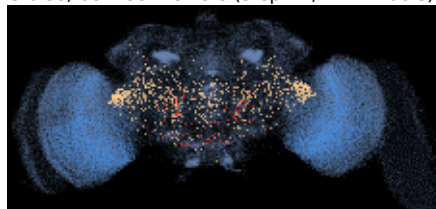
● other ● visual ● olfactory ● taste ● DN ● motor ● all other neurons (not firing)

a Perch (step -9)



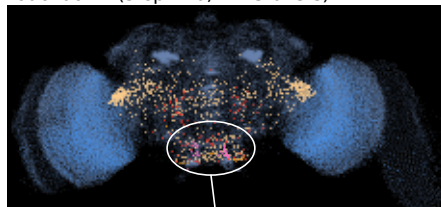
11,920 neurons fired in this 25 ms step
network mean 6.6 Hz

b Cruise, between towers (step 47, $t = 1.200$ s)



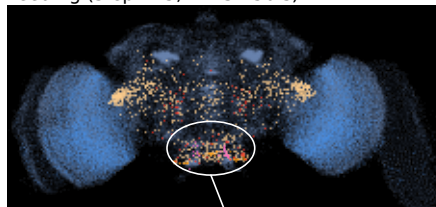
12,709 neurons fired in this 25 ms step
network mean 7.3 Hz

c Touchdown (step 120, $t = 3.025$ s)

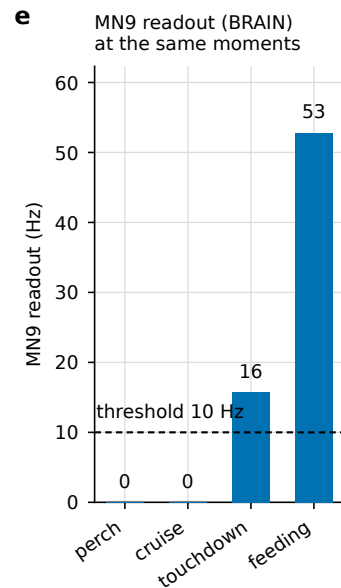


taste GRNs and motor neurons fire after
touchdown (cruise → touchdown):
taste 0 → 34, motor 2 → 14 neurons per step
13,817 neurons fired in this 25 ms step
network mean 8.1 Hz

d Feeding (step 125, $t = 3.150$ s)



taste GRNs and motor neurons fire after
touchdown (touchdown → feeding):
taste 34 → 31, motor 14 → 25 neurons per step
13,331 neurons fired in this 25 ms step
network mean 7.9 Hz



Display settings, not measurements: dot position (arbor centroid of each neuron, frontal view; the fly's left is on the right), class colour, dot size (equal for all classes), dot opacity (visual 13 %, other classes 85 %), drawing order and the dim background cloud. Measured in the model: which neurons fired in the 25 ms step, their number, the network mean rate, the MN9 readout (perch: recorded perch mean; step -9 = last perch step before the input cut). Run: n1, seed 3. Visual input comes from FlyVis (FLYVIS); the route is hand-made (HAND-MADE); the only behaviour the brain controls is the feeding decision (BRAIN: MN9 > 10 Hz).